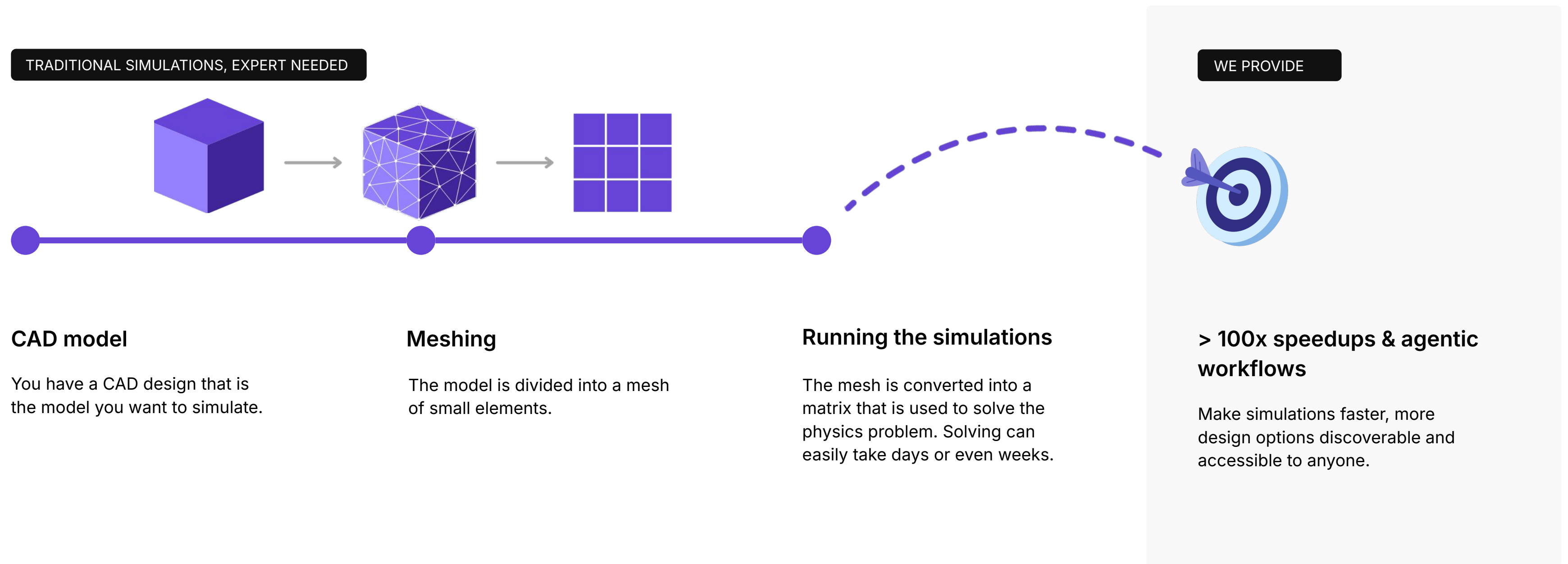


Quanscient Allsolve
Vibe-code a real physics simulation
app

Mila Huovinen
AI Developer Intern

FEM simulations

Behind every hardware product



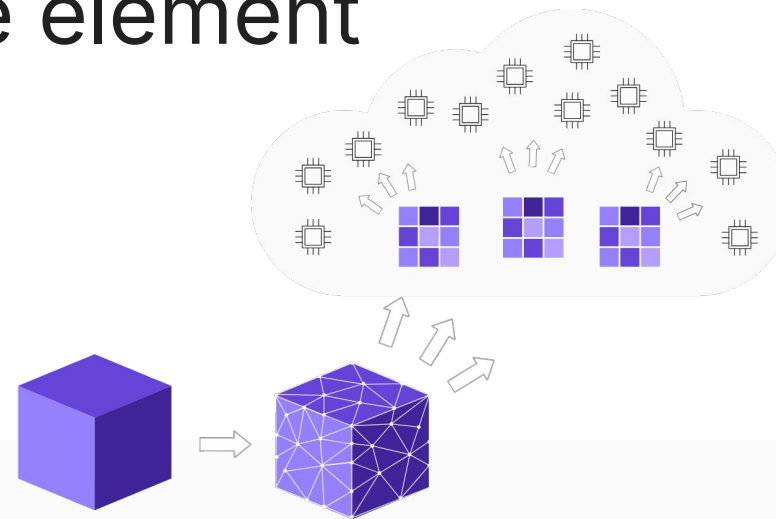
Our product: Quanscient Allsolve FEM Multiphysics simulations in the Cloud

We provide **speed, volume, and automation** for multiphysics and multiscale finite element simulations



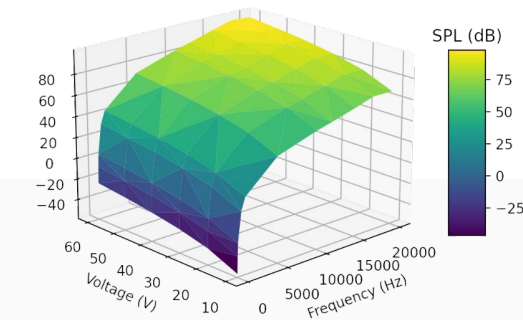
Allsolve UI

- Web app
- See visually what's happening with the model
- Allsolve Agent to help you



Allsolve SDK

- Way to connect Allsolve from Python code
- Let (for example Cursor) AI agents write and run your simulation scripts without clicking the UI



Your own tool (that you will make today)

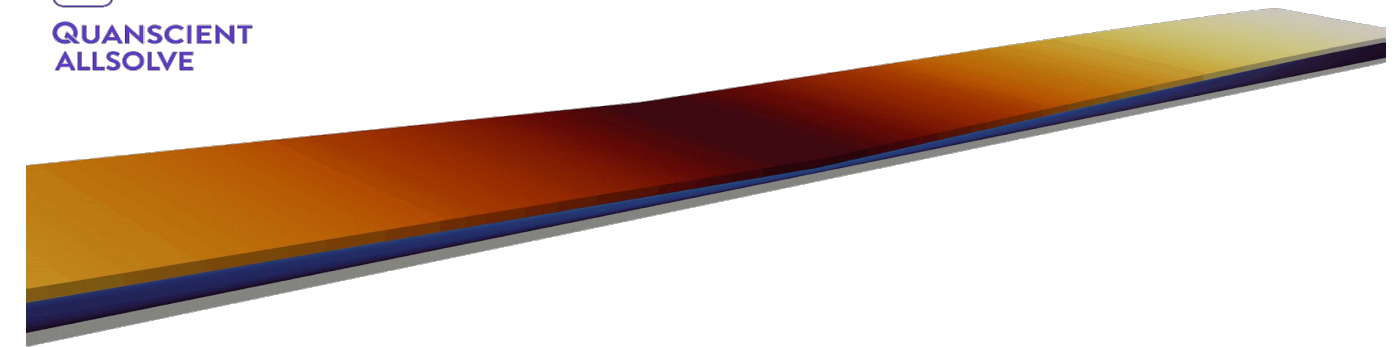
- Connecting to Allsolve through the provided SDK
- Solves some real life problem

Your challenge

Pick a real engineering problem. Build an app that solves it with Allsolve simulations.



QUANSCIENT
ALLSOLVE



Why this challenge matters

- Engineering simulation usually means one expert working with complex desktop software.
- Quanscient Allsolve SDK exposes that same simulation as code, so anyone can build a custom app for a specific problem. AI coding tools make this a few hours' work instead of months.
- We want to see what our SDK can enable: new use cases, interfaces, and ways to solve and optimize real engineering problems faster.

78%

Of engineers **need to reduce complexity** of their simulations due to lack of computational capacity.

→ **Time consuming and a huge compromise in accuracy.**

96%

Of engineering decision makers say **reducing time, risk, and cost** is the most valuable in their R&D.

→ **Lacking simulation capabilities lead to unnecessary prototyping.**



Beer Cooling Simulator

Powered by Quanscient Allsolve

quanscient.com →

Simulation Parameters

Container Type

 can

 bottle

 pint

Cooling Method

 Ice Water Bath (0°C) ▼

Initial Temperature

20°C

-10°C

Room temp (20°C)

40°C

Target Temperature

4°C

-10°C

Perfect drinking (4°C)

20°C

Immersion Level

80%

0% (floating)

100% (fully submerged)

Simulation Duration

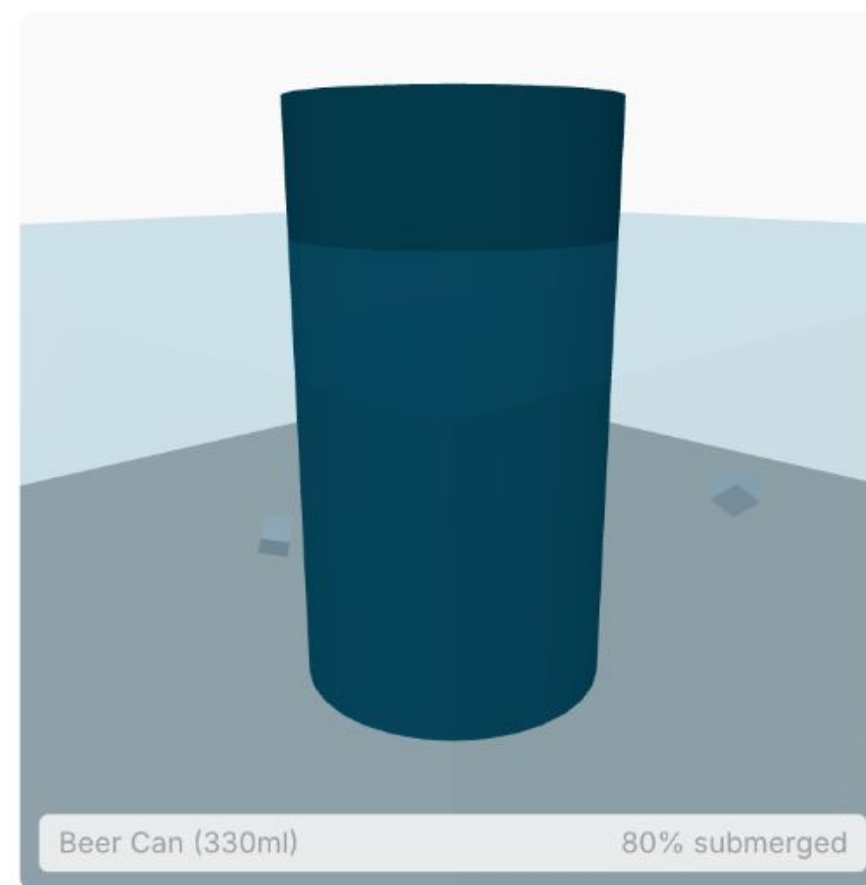
30 min

 Allsolve Cloud

Real FEM simulation

Runs on Quanscient Allsolve cloud (requires API key)

 Start Cooling



CURRENT TEMPERATURE

0.7°C

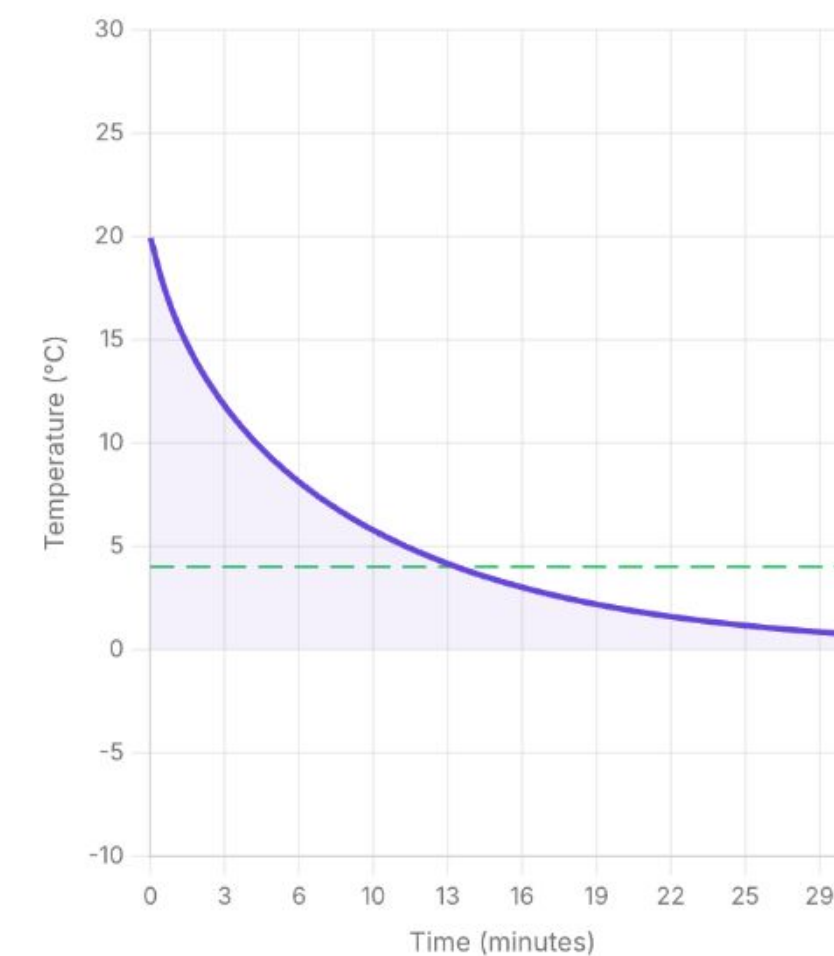
Time: 30:00

✓ Target: 4°C (reached in 13:10)

Final Temp
0.7°C

Time to Target
13:10

Temperature vs Time



Your challenge

Pick an example customer with an engineering problem. Show how an Allsolve-powered app creates value for them.

A great solution looks like

- A working app that takes user input, calls Allsolve via the SDK, runs a real simulation and returns a result the user can act on.
- The solution should address a realistic and meaningful use case. Show the real industry problem and how your app creates value for the user instead of using just our UI.

Winning criteria

1. A meaningful industry problem, which provides value for the user
2. Creativity / originality of the use case
3. SDK experience and insights; how clearly solution identifies SDK usability, pain points, and opportunities for improvement
4. A working demo, physics actually used and logic seems correct

We'll provide:

- Access to Quanscient Allsolve
- Agentic skills to help access Allsolve through Cursor
- Support during the challenge

Useful links:

- <https://allsolve.quanscient.com/>
- <https://allsolve.quanscient.com/documentation>
- <https://github.com/Quanscient-Public/allsolve-sdk-python>
- `pip install allsolve`



Quanscient Allsolve Docs

Learn about Quanscient Allsolve's features, how to use them, and start simulating!

Get started →

Thank you!
Join the new era in simulations.

QUANSCIENT

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